

Syllabus

Lecture:

Mon, Wed, Fri
3:00pm–3:50pm
Location: Spieth 2200

Instructor:

Christian Shelton
(cshelton@cs.ucr.edu)
Office hrs: Thu 4:00pm–6:00pm
Location: MRB 4118

Discussion:

Tue 1:00pm–1:50pm
ND A1010
TA: Xiaoyin Pan
(xiaoyin.pan@email.ucr.edu)

Required Text: none Preprints of a textbook will be provided. *Please* do not redistribute these.

Course Purpose: Introduce the methods, ideas, and techniques in machine learning and data mining.

Prerequisites: Linear Algebra (such as MATH 31 or EE 20), Multivariate Calculus (such as MATH 10A), Intro Probability/Statistics (such as STAT 155 or EE 114), and solid programming (such as CS 100 or EE 120B). These prerequisites have been very carefully considered. If you do not have any one of them, you will need to first complete them, and *then* take this course. This course is currently offered every quarter.

Instruction Mode: This course will be delivered live, in person. Lecture and discussion attendance is critical to keeping pace with this course.

Messages, discussions, resources, and grades will be on <https://elearn.ucr.edu/> (Canvas). **You must keep current with the announcements and discussions on eLearn, as changes will be announced there.** We will also monitor these message boards closely, so please use them to ask questions outside of lecture.

Class Schedule:

Monday	Wednesday	Friday	Due
3/30 Introduction [ch 1,2]	4/1 Bayes-optimal [ch 3]	4/3 Naive Bayes [ch 3]	
4/6 Linear regression [ch 4]	4/8 Linear regression [ch 4]	4/10 Perceptron [ch 5]	PS 1
4/13 Logistic Regression [ch 5]	4/15 Logistic Regression [ch 5]	4/17 Features [ch 6]	Q1 [3–5]
4/20 Regularization [ch 6]	4/22 Validation [ch 6]	4/24 SVM [ch 7]	PS 2
4/27 SVM [ch 7]	4/29 Kernels [ch 7]	5/1 Nearest Neighbor [ch 9]	Q2 [6–7]
5/4 Normalization [ch 9]	5/6 Neural Networks [ch 8]	5/8 Neural Networks [ch 8]	PS 3
5/11 Neural Networks [ch 8]	5/13 Decision Trees [ch 10]	5/15 Decision Trees [ch 10]	Q3 [8–9]
5/18 Decision Trees [ch 10]	5/20 Clustering [ch 11]	5/22 Clustering [ch 11]	PS 4
5/25 <i>Memorial Day</i>	5/27 Feature Selection [ch 12]	5/29 PCA [ch 12]	Q4 [10–12]
6/1 PCA [ch 12]	6/3 Data Representations [ch 13]	6/5 Review	
————— Final Exam: Monday, June 8, 8:00am–11:00am —————			

Course Work: Four **problem sets** will be due during the quarter, each due on Friday evening. They will be a mix of written-response and programming questions, in Python. They are non-trivial, so allocate sufficient time for them. The problem sets will be distributed, completed, collected, and returned on the CS department’s Jupyter Hub server at <https://hub.cs.ucr.edu/>. Instructions and an example will be provided.

Four **quizzes** (50 minutes) and one **final exam** (3 hours) are scheduled for this course. The quizzes each cover the material since the previous quiz. The final covers all material in the course. The quizzes will be taken in the CS testing center and must be scheduled during Friday of the week shown above. Because this is my first time using the testing center, I reserve the right to adjust the format and location of the tests. The final exam may use the testing center, depending on availability.

Late Work: Each student is granted four “late days” which can be used (in integer units) on any of the problem sets. If a more dire situation arises, please contact the instructor. To use a late day, just submit the problem set late; there is no need to inform the instructors first.

External sources: For the problem sets, you may consult webpages that explain the Python programming language, *provided they do not cover any machine learning topics*. Good resources are <https://docs.python.org/3/> for Python in general and <https://docs.scipy.org/doc/numpy/> for the numpy package. You may also reference textbooks, course slides, and course lectures.

Generative AI (ChatGPT and the like) may be used only in a similar fashion: to query background material like Python or linear algebra. Entering any specifics of the problem sets or querying things specific to the problem

set is not allowed. If in doubt, **please just ask**.

Cheating: I will not tolerate any kind of cheating in this course. Problem sets and exams are to be completed on your own. The only external sources allowed are those mentioned above or by the instructor throughout the course. **If you have a doubt or question, please just ASK.**

You may not solicit, reference, or copy any material not directly sanctioned as above. You may not provide your solutions or the supplied materials of this course to anyone else.

You may not use generative AI (ChatGPT and the like) for any part of any assignment or exam except as detailed above.

While you may discuss the class material with other students, that discussion may *not* include direct reference to any solutions. As examples, you may ask and answer questions such as “Should I plot both values on the same plot or different plots?” and “Does agglomerative clustering with complete linkage combine the clusters with the maximum or minimum distances?” But, you may not ask or answer questions such as “What accuracy did you get on part a?” and “How did you implement the gradient update rule?” If you are looking at code for an assignment with someone else, that is almost certainly a violation.

As per standard UCR policy, you may not submit answers (written or programming) to problem sets that contain material you did not produce yourself for the express purpose of *this* offering of this course. If I find that you have submitted work that is not your own or is work you submitted in a different course, I will assign you a zero on that assignment (and possibly a zero on the entire course, depending on the severity), and I will forward the case to Student Conduct & Academic Integrity Programs for campus-level consideration.

Grade: The weights of each graded component are

Problem sets	10% each
Quizzes	10% each
Final	20%

Grades will be assigned based on final total score, using standard “90%-80%-70%-60%” cut-offs for letter grades. However, this will be done *after* the denominator (total number of points in the course) has been reduced to account for questions that were found after grading to be too difficult. *Typically* this reduces the denominator by about 10% (which has the effect of lowering the effective cut-offs to “81%-72%-63%-54%”).

There is no extra credit available in this course.

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Similarly, you own the copyright on your original problem set solutions, code, and exams. If I wish to post such answers, I must secure your written permission.